

Name: __Dog Paw Washer_____

Date: _3/12/2026_____

Class: __pd 2_____

MyDesign Portfolio

Element C Initial Template -AT - 18 points

How to use this template

Element C: Presentation and Justification of Solution Design Requirements

Design Requirement #1 (HIGH Priority)

The prototype must mechanically remove visible dirt and debris from a dog's paw using a contained cleaning mechanism without causing any liquid it uses to spill outside the device.

This requirement ties directly to the team's stated goals of cleaning paws without spilling liquid onto the floor. It also validates the product's overall goal of preventing mess, so not only will it prevent the mess caused by the dog, but it itself won't cause a mess.

Primary stakeholders, including dog owners among the team members, validated this requirement and agreed with the goal to prevent mess.

Design Requirement #2 (HIGH Priority)

The prototype must dry the dog's paw before it leaves the device using an integrated drying method.

This requirement directly ties to the team's stated goal of ensuring the dog's paw is dry before exiting the device. If the paw is not dry after leaving the device, then contaminated water will be tracked into the home, and further cause a mess.

Stakeholders, including dog owners within the team, confirmed this requirement because wet paws would still create a mess even if dirt is removed, so drying is a requirement.

Design Requirement #3 (HIGH Priority)

The prototype must be structurally robust and capable of resisting tipping or damage under vertical force from small to medium-sized dogs.

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This requirement ties directly to the team's definition of success, which includes structural robustness and resistance to excessive weight and vertical force. The device has to be able to accommodate dogs of varying sizes, so structural strength is integral to the product's functionality.

Stakeholders validated this requirement because a device placed in a doorway must withstand repeated use and not be easily knocked over.

Design Requirement #4 (HIGH Priority)

The final prototype must be portable.

Previously cited as a requirement, it meets the needs of dog owners who may have multiple dogs or multiple entrances to their homes. If the device is portable, cleaning a dog or other pet can be a valid first step before entering the home from any entrance. The device can also be taken on trips if owners like to travel with their pets.

Primary stakeholders, including dog owners within the team, approved this design requirement, recognizing the importance of portability as dogs and their owners move around.

Design Requirement #5 (HIGH Priority)

Our initial prototypes must be fully assembled and operational using only classroom-available materials which include but are not limited to cardboard, wood, plastics, 3D printed parts, string, tape, glue, and other approved materials with no external purchases.

This directly aligns with the project's triple constraint of cost and material limitations outlined in the team charter.

Primary stakeholders which include dog owners among team members approve this requirement, especially considering cost constraints and timelines.

Design Requirement #6 (LOW Priority)

The prototype should complete the full cleaning and drying process for one paw in under 30 seconds to ensure practical usability.

A faster process improves real-world functionality in a residential entryway setting, making the device more convenient for dog owners. Generally, for routine walks or bathroom breaks,

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owners are taking time out of their busy days to attend to their pets' needs. A speedy cleaning process helps alleviate some of the time consumption of walking a dog.

Primary stakeholders approved this requirement, and even recommended it be higher up in terms of project importance, citing the necessity for reliable and fast functionality.

Design Requirement #7 (LOW Priority)

This prototype should have an aesthetically clean and organized exterior design suitable for placement in a residential mudroom or garage entryway.

Because the device is intended for visible placement in a home entryway, appearance contributes to practical adoption and usability. Something that is glaring or visually unappealing is less likely to be adopted or consumed by customers, so creating a device that looks good ensures that users are comfortable with it in their home.

Primary stakeholders, including the parents of dog owners who are team members, approved this requirement, citing that the device must be visually pleasing, taking up space in important entryways.